

ICT102 Assignment: Recursion (Factorial & Fibonacci)

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This assignment demonstrates the use of recursion in solving two common problems: factorial and Fibonacci sequence. The solutions include flowcharts, pseudocode, and Python implementations.

Flowcharts

Flowchart Description (Factorial):

1. Start
2. Input number n
3. If n is negative → display error
4. If n equals 0 → return 1
5. Otherwise → return $n \times \text{factorial}(n-1)$
6. End

Flowchart Description (Fibonacci):

1. Start
2. Input number n
3. If n is negative → display error
4. If n equals 0 → return 0
5. If n equals 1 → return 1
6. Otherwise → return $\text{fibonacci}(n-1) + \text{fibonacci}(n-2)$
7. End

Pseudocode

Factorial:

```
BEGIN
  IF n < 0 THEN
    PRINT "Invalid input"
  ELSE IF n == 0 THEN
    RETURN 1
  ELSE
    RETURN n × factorial(n - 1)
END
```

Fibonacci:

```
BEGIN
  IF n < 0 THEN
    PRINT "Invalid input"
  ELSE IF n == 0 THEN
    RETURN 0
  ELSE IF n == 1 THEN
    RETURN 1
  ELSE
    RETURN fibonacci(n - 1) + fibonacci(n - 2)
END
```

Python Implementation

```
def factorial(n):
    if n < 0:
        return "Invalid input"
    if n == 0:
        return 1
    return n * factorial(n - 1)

def fibonacci(n):
    if n < 0:
        return "Invalid input"
    if n == 0:
        return 0
```

```
        if n == 1:
            return 1
        return fibonacci(n - 1) + fibonacci(n - 2)

# main program
try:
    number = int(input("Enter a number: "))
    print("Factorial:", factorial(number))
    print("Fibonacci:", fibonacci(number))
except:
    print("Invalid input. Please enter an integer.")
```